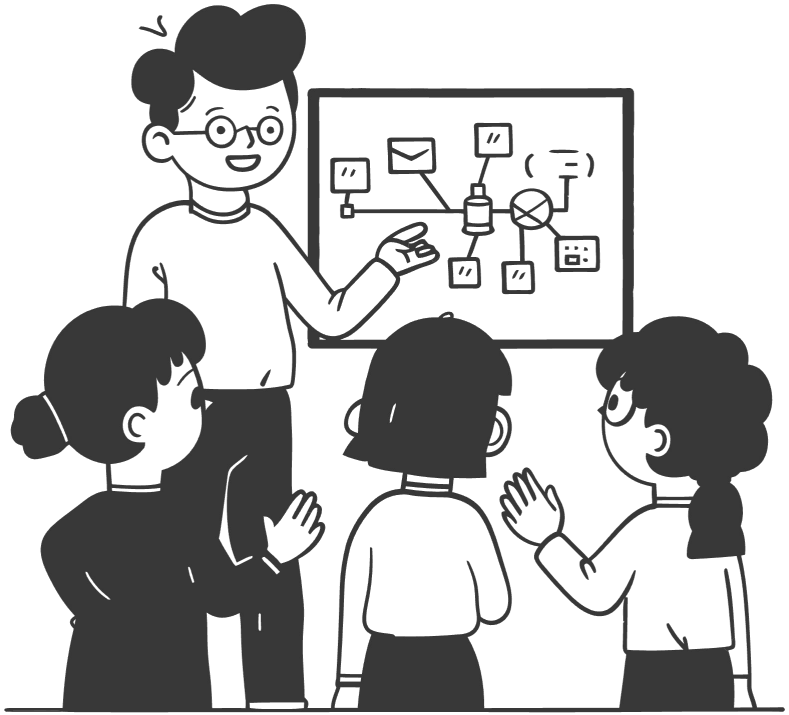
The DATA-MD Curriculum

Session 4 – AI Documentation and Future Directions

Data-Driven, Technology-Assisted Medical Decision Making for Medical Trainees

- Emily Balczewski, PhD · Andrew Wong, MD, MS · Cornelius James, MD
- Niharika Goparaju, MD · Thomas East, PhD · Joanne Baker, DO
- Thalia Nguyen, MD · Serena Wang, MD · Lisa Zickuhr, MD





Course Outline

Session 1: Introduction to AI

What is AI? Why do we care?

Session 2: Error and Bias in AI Systems

AI for Clinical Decision Making

Session 3: Evaluating and Appraising AI Systems

AI Benchmarks; Critical Appraisal

Session 4: Clinical Applications of AI

AI for Documentation; Future Directions

Session 4 Learning Objectives

- **Critically appraise**

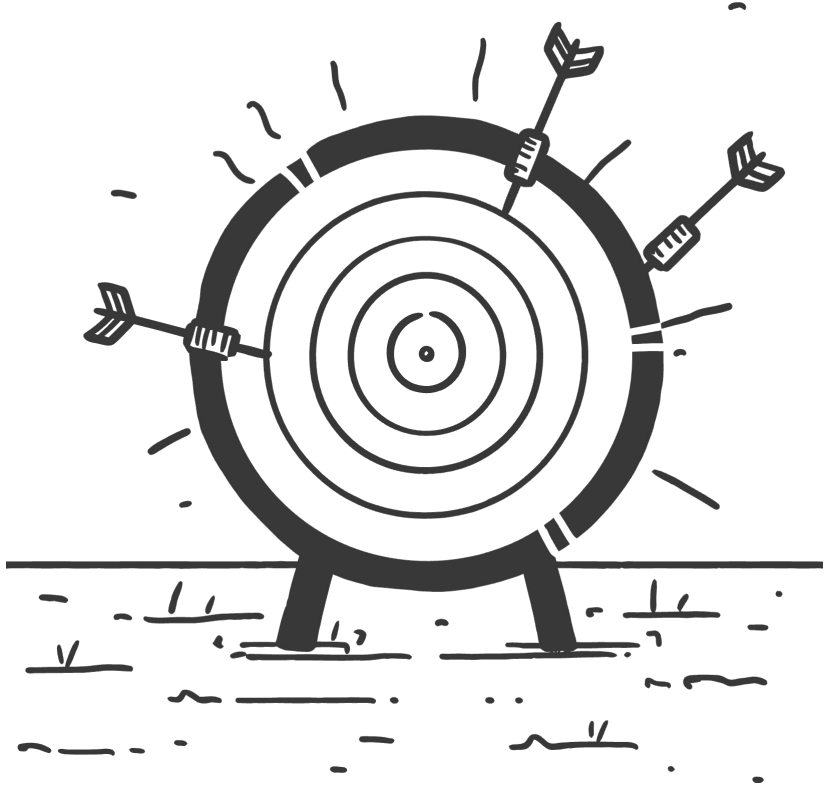
  Evaluate AI tools for clinical documentation

- **Identify resources**

  Locate sources for AI learning

- **Discuss future directions**

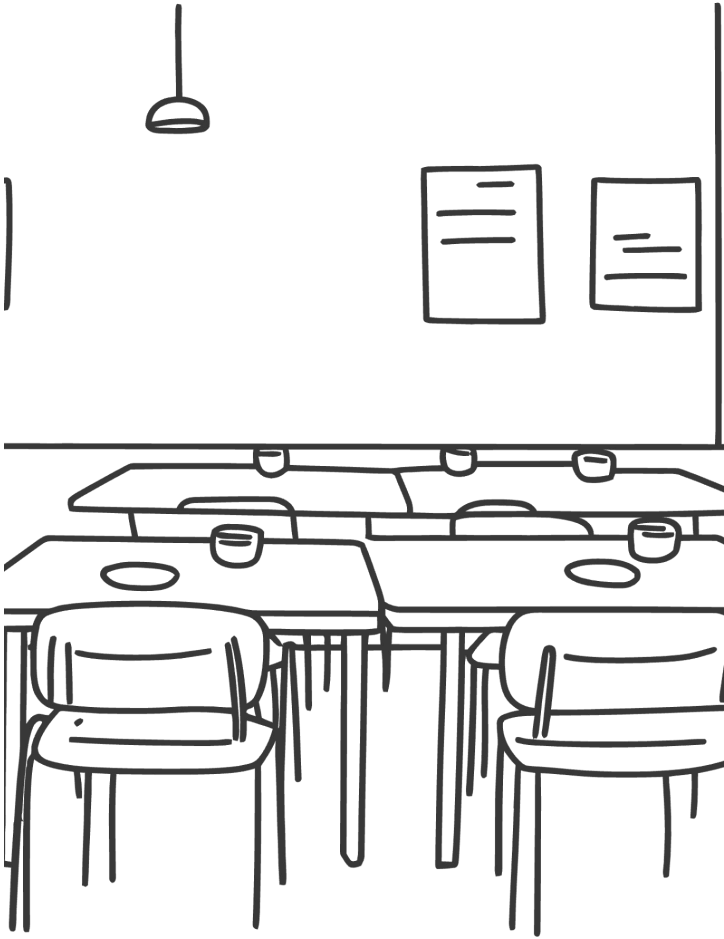
  Project AI performance and workforce impacts



REVIEW

Session 3 Review

Recapping key concepts from **Evaluating and Appraising AI Systems**: benchmarks, critical appraisal, and levels of automation.



Common Generative AI Sources of Error

1: Hallucination

Fictional, erroneous, or unsubstantiated information presented as fact

a) Inaccuracy: Correct structure, but one or more details are factually wrong

b) Misattribution: Wrong source cited, or source misinterpreted

2: Omission

Clinically important information missing from the output

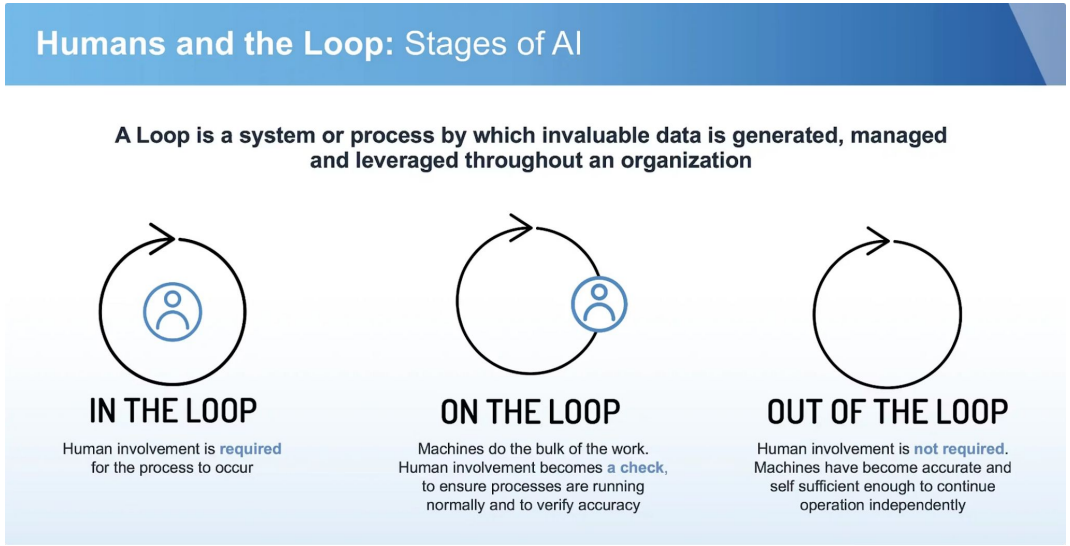
3: Limited Reasoning

Draws incorrect conclusions even from correct premises

4: Text Output Errors

Spelling, translation, grammar, or fluency mistakes

Humans and the Loop



In the Loop

Human involvement is **required** for the process to occur

On the Loop

Machines do the bulk of work; humans **check and verify**

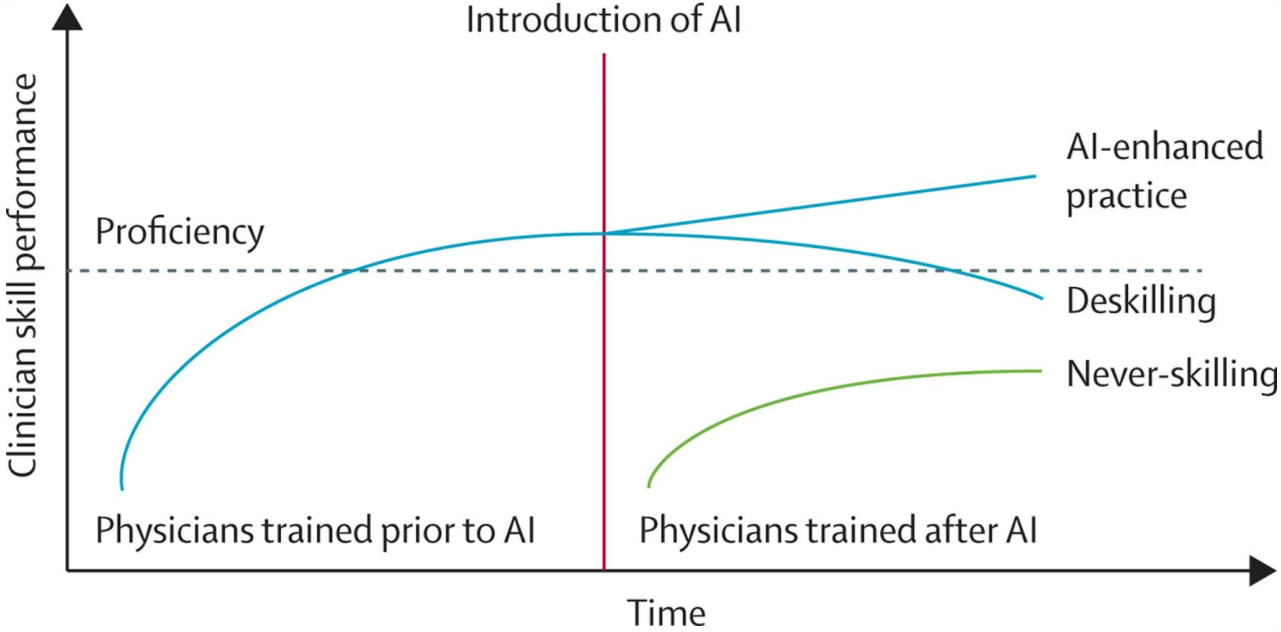
Out of the Loop

Human involvement is **not required** — machines operate independently

Parasuraman, IEEE Transactions on Systems, Man, and Cybernetics (2000)

Deskilling and Never-Skilling

A key drawback of automation: clinicians who rely on AI may lose — or never develop — core skills.



Deskilling

Physicians trained before AI: proficiency declines with over-reliance

Never-Skilling

Physicians trained after AI: may never reach baseline proficiency

AI-Enhanced Practice

Optimal path: AI augments rather than replaces clinical judgment

How To Read An AI Paper In Under 15 Minutes: A PEM Pocket Framework

1. What clinical task is the model doing?

If you cannot name the task in one sentence, the paper is not ready to translate to your shop.

2. Efficacy or effectiveness (clinical deployment/effective trials)?

If efficacy only, you cannot conclude the tool will help your patients - only that it could.

3. What was the comparator?

Many AI papers benchmark against the weakest plausible human comparator and then headline “AI matches doctors.” Read the comparator carefully.

4. What was the prompt or benchmark?

Was the model spoon-fed a clean, complete vignette - or did it have to gather information itself, the way you do at the bedside? Performance usually drops sharply when the model has to ask for the data. The benchmark world is not the ED.

5. Who's missing from the data?

Pediatric patients are often invisible in adult-trained models, and tools that look great in adults can fail silently on neonates and infants.

6. What kinds of errors does the model make, and are they harmful?

A model that confidently produces extreme errors with no way to know it is wrong is the worst case.

Bonus question for vendor pitches

If a vendor walks into your group meeting next month with this tool, what specific evidence would you need to see before letting them deploy on your patients?

Critical Appraisal #1 — Key Findings

SCTs as Benchmarks

Script concordance tests may be more valid than MCQs for assessing clinical reasoning

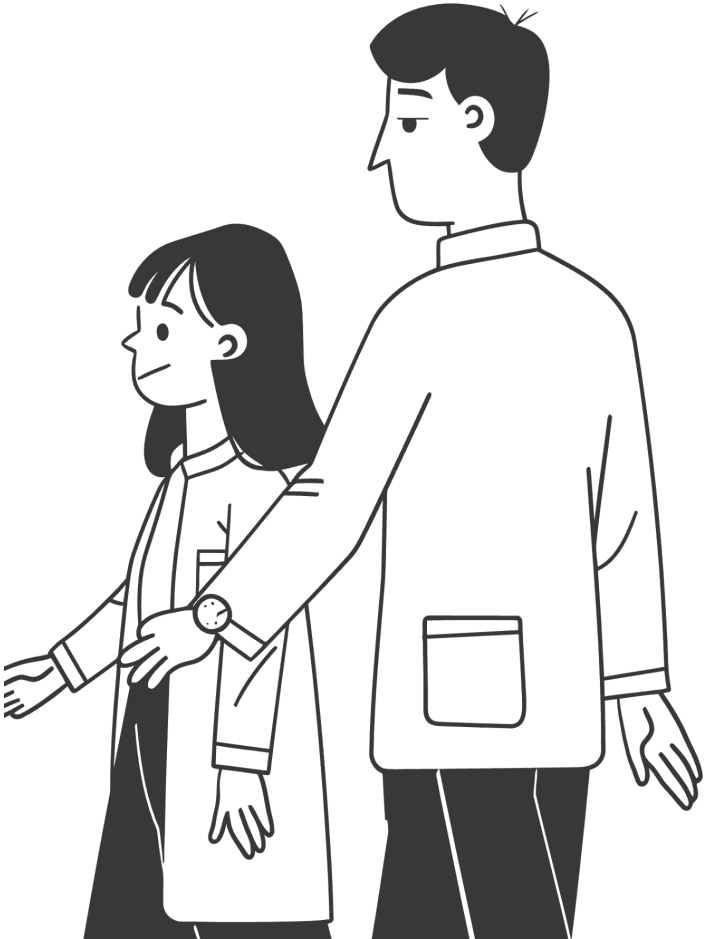
Model Performance

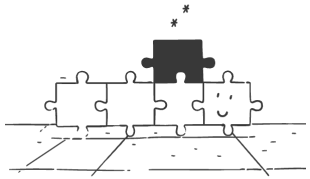
Models/prompts performed similarly to each other; outperformed students but **not** senior residents or faculty

Reasoning Patterns

Models showed a distinctly different pattern of "reasoning" compared to human clinicians

⚠ **Key Gap:** No RCTs of LLMs used for clinical reasoning in a real clinical context (efficacy vs. effectiveness)







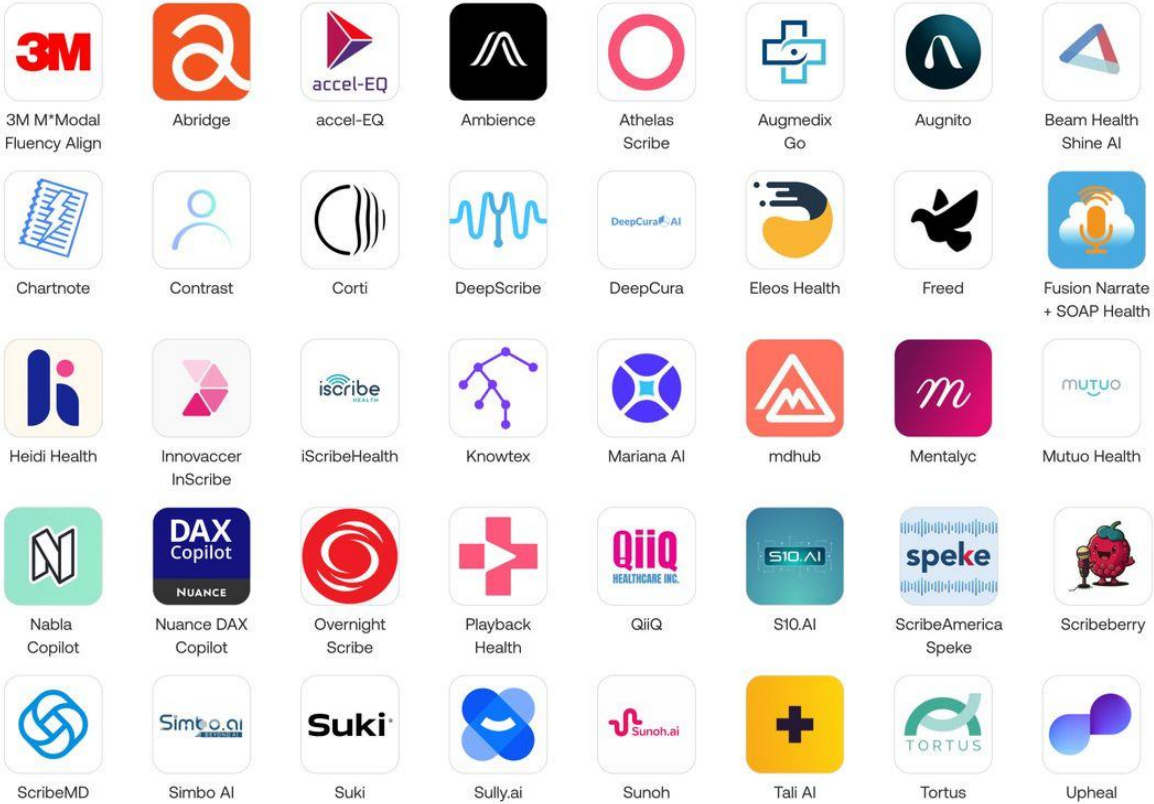
NEW TOPIC

Background for Critical Appraisal #2

Setting the stage for evaluating AI documentation tools in clinical practice.

Elion

AI Scribe Category



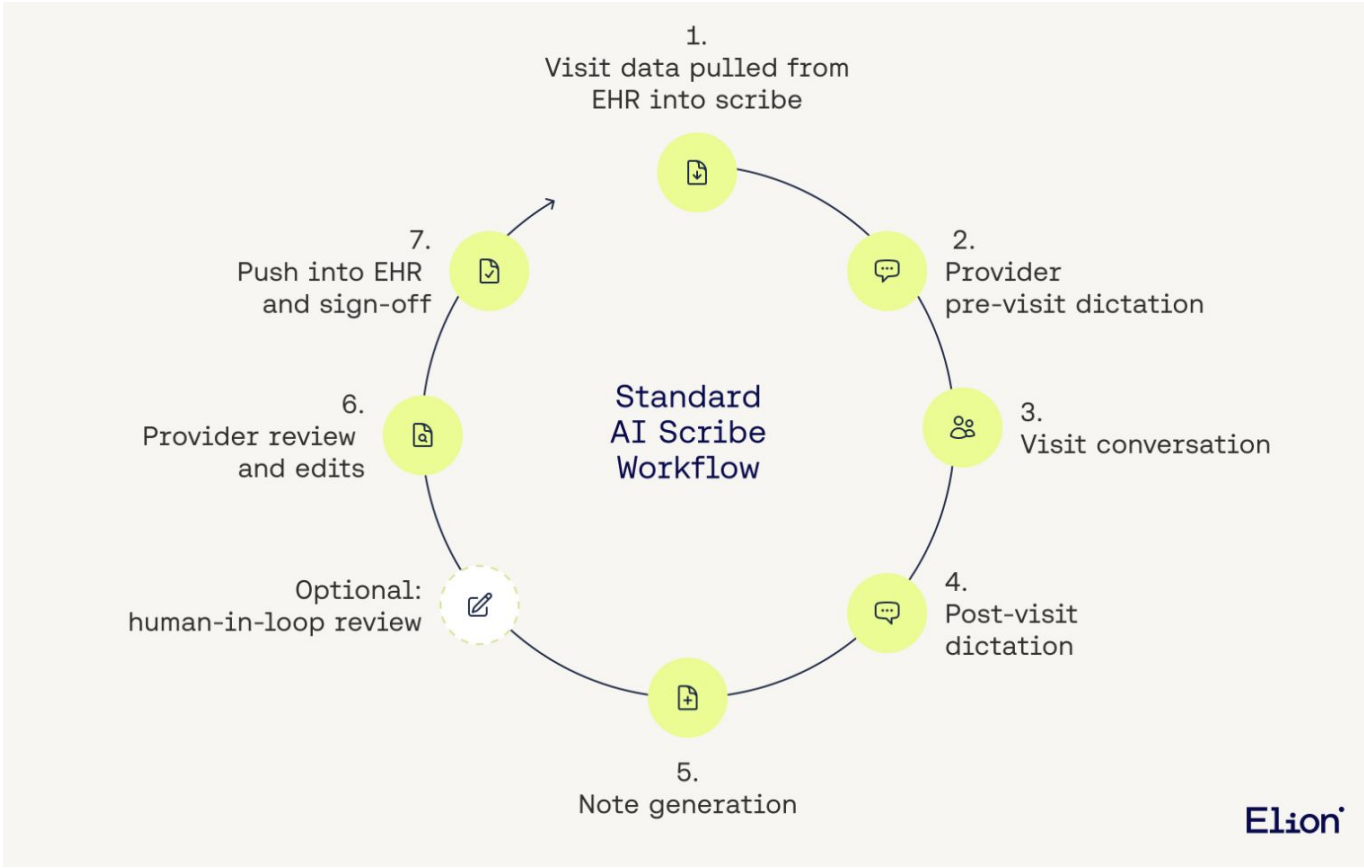

1.
Visit data pulled from
EHR into scribe
2.
Provider
pre-visit dictation
3.
Visit conversation
4.
Post-visit
dictation
5.
Note generation
Optional:
human-in-loop review
6.
Provider review
and edits
7.
Push into EHR
and sign-off
Standard
AI Scribe
Workflow
Elion

Exhibit 10

METRICS FOR EVALUATING THE IMPACT OF AMBIENT SCRIBES



Process Metrics

- Time in notes
- Time in EHR
- Percentage adoption (specialty, overall)
- Percentage encounters with ambient (specialty, overall, visit type)
- Number of provider encounters with ambient
- Time to note closure
- Percentage of notes closed same day
- Time in visit

Experience Metrics

- Clinician burnout (e.g., custom surveys, Mini Z survey, Stanford Model of Professional Fulfillment, Copenhagen Burnout Inventory)
- Clinician turnover
- Technology retention rate
- "Pajama time" (time spent on documentation after work)
- Usability and clinician experience surveys
- Patient experience (e.g., custom surveys, CH-CAHPS, Press Ganey)

Financial Metrics

- Relative Value Unit (RVU) per given period of time
- Number of patient encounters per period of time
- Average E&M code complexity level
- HCC capture and accuracy

Quality Metrics

- Percentage of note kept
- Developer-led documentation quality scores
- Word error rate
- Medical term recall rate

JOURNAL ARTICLE FEATURED

Ambient artificial intelligence scribes: physician burnout and perspectives on usability and documentation burden

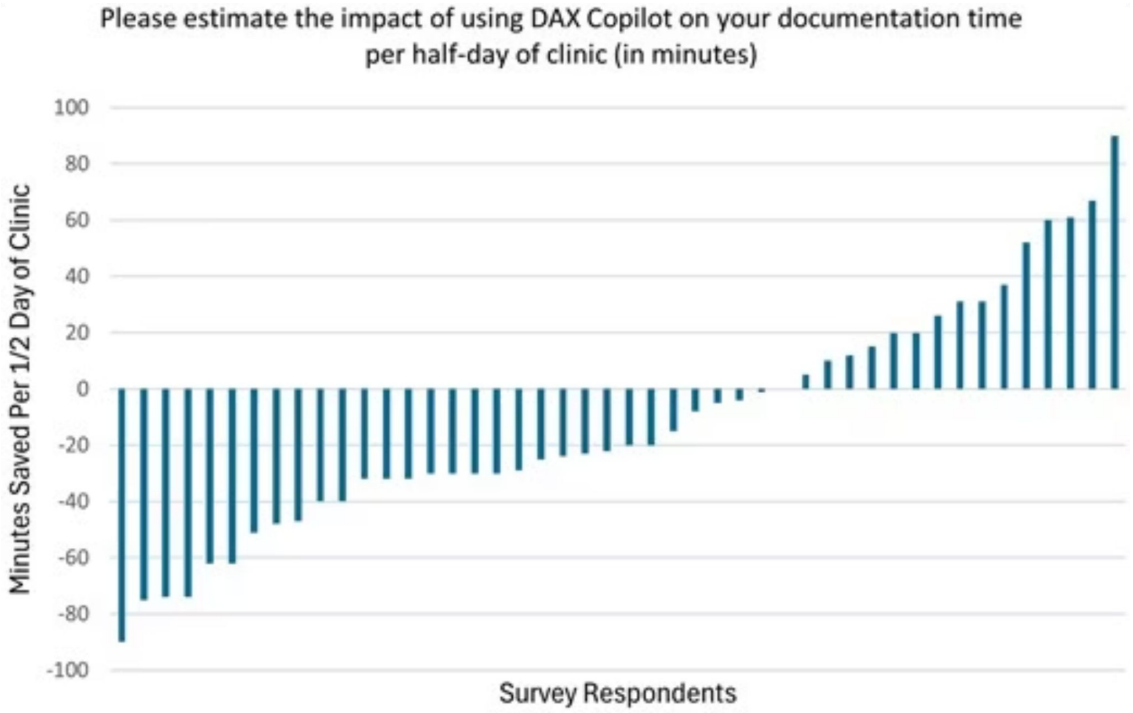
Shreya J Shah, MD ✉, Anna Devon-Sand, MPH, Stephen P Ma, MD, PhD, Yejin Jeong, BA, Trevor Crowell, BA, Margaret Smith, MBA, April S Liang, MD, Clarissa Delahaie, BAS, Caroline Hsia, MEng, Tait Shanafelt, MD ...
Show more

Journal of the American Medical Informatics Association, Volume 32, Issue 2, February 2025, Pages 375–380,
https://doi.org/10.1093/jamia/ocae295

Published: 05 December 2024 **Article history** ▾

Topic	Published	Key Question
Ambient AI scribes, physician burnout, usability, and documentation burden	*JAMIA*, Volume 32, Issue 2 — February 2025	Does ambient AI documentation reduce burden or create new challenges?

DAX Copilot: Documentation Time Impact



Concerns:

1. Accuracy
2. Liability risks (privacy violations)
3. Loss of control of clinical narrative
    a. “algorithmic homogenization”

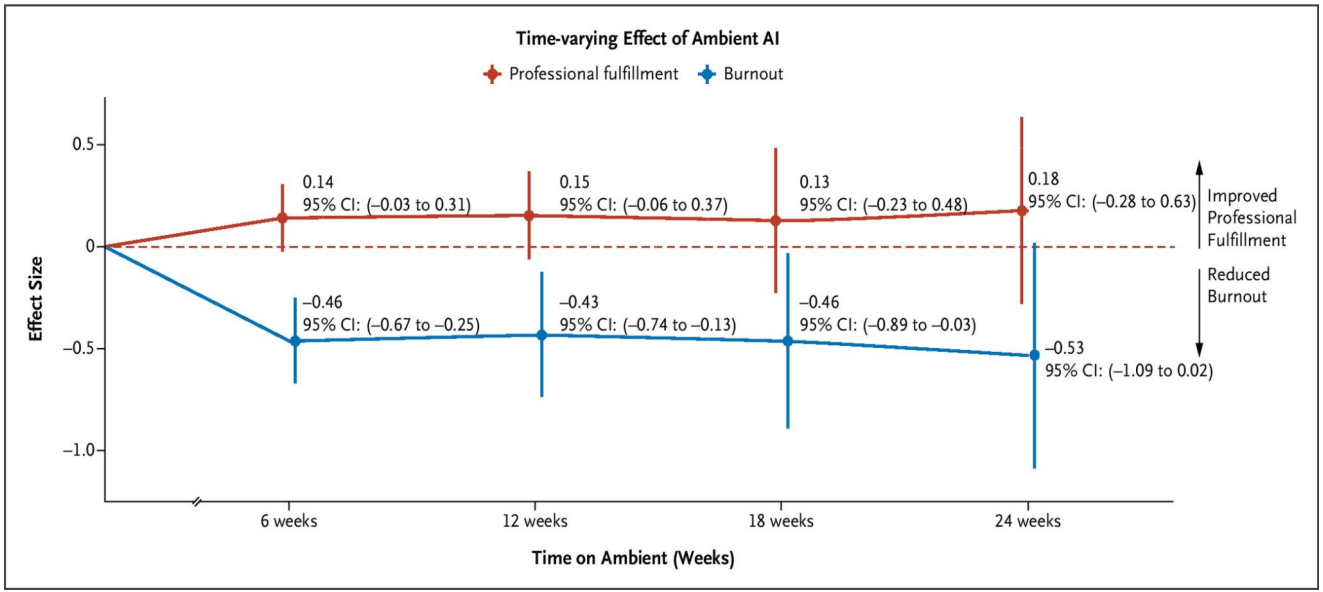
Physician-reported minutes saved (or added) per half-day of clinic using DAX Copilot. Note the **wide variability** — some physicians saved significant time, while others lost time.

A Pragmatic Randomized Controlled Trial of Ambient Artificial Intelligence to Improve Health Practitioner Well-Being

Authors: Majid Afshar, M.D., M.S. , Mary Ryan Baumann, Ph.D. , Felice Resnik, Ph.D. , Josie Hintzke, M.S. , Anne Gravel Sullivan, Ph.D. , Graham Wills, Ph.D. , Kayla Lemmon, M.S. , +14 , and Joel Gordon, M.D. Author Info & Affiliations

Published November 26, 2025 | NEJM AI 2025;2(12) | DOI: 10.1056/AIoa2500945 | VOL. 2 NO. 12

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JAMA Internal Medicine | Original Investigation

Physician- and Large Language Model–Generated Hospital Discharge Summaries

Christopher Y. K. Williams, MB BChir; Charumathi Raghu Subramanian, MD; Syed Salman Ali, MD; Michael Apolinario, MD; Elisabeth Askin, MD; Peter Barish, MD; Monica Cheng, MD; W. James Deardorff, MD; Nisha Donthi, MD; Smitha Ganeshan, MD, MBA; Owen Huang, MD; Molly A. Kantor, MD; Andrew R. Lai, MD, MPH; Ashley Manchanda, DO; Kendra A. Moore, MD, MBE; Anoop N. Muniyappa, MD, MS; Geethu Nair, MD; Prashant P. Patel, DO; Lekshmi Santhosh, MD, MA Ed; Susan Schneider, MD, MSPH; Shawn Torres, MD; Michi Yukawa, MD, MPH; Colin C. Hubbard, PhD; Benjamin I. Rosner, MD, PhD

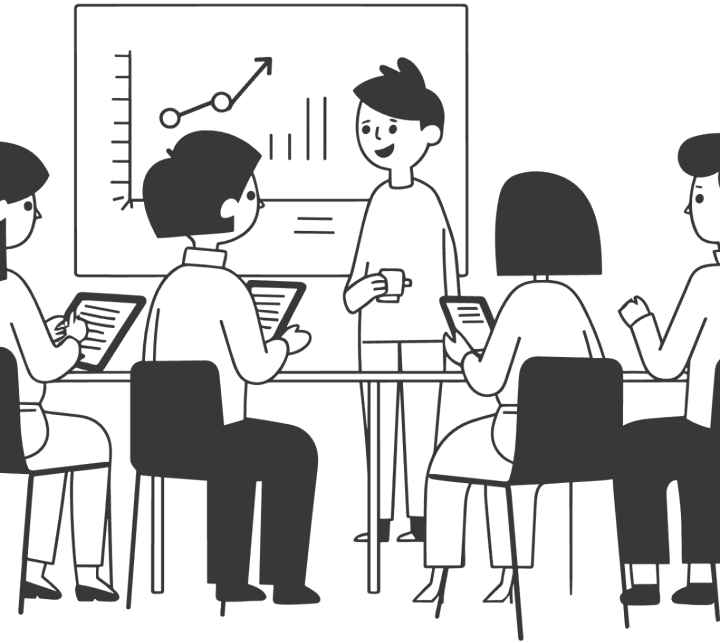
CRITICAL APPRAISAL

Critical Appraisal: Documentation

Evaluating LLM-generated vs. physician-generated clinical notes: quality, safety, and practical implications.



QR ME for the PDF of the STUDY



Discussion Outline

01

Abstract *(5 min)*

Read and discuss aims and key results

02

Introduction *(5 min)*

Identify the knowledge gap and study rationale

03

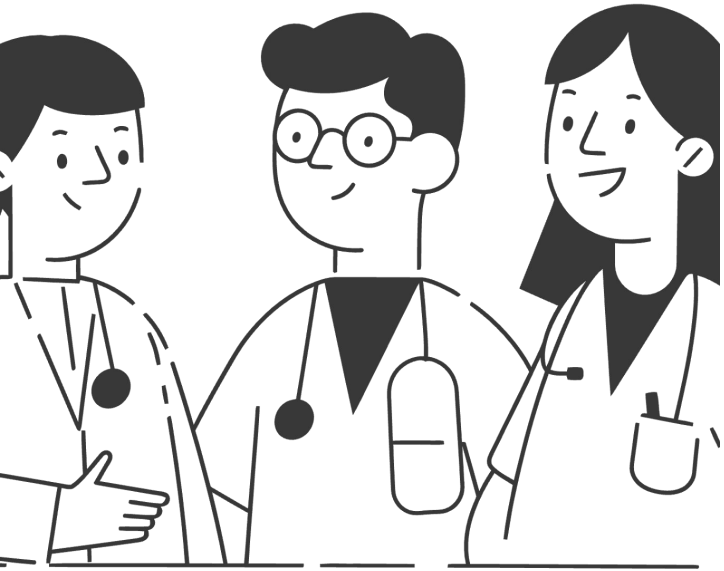
Methods *(10 min)*

Evaluate prompts, models, inclusion/exclusion criteria

04

Results & Conclusions *(10 min)*

Interpret quality measures, harm data, and automation implications



Introduction - Knowledge Gap

Why Good Discharge Summaries Matter

- Reduce hospital readmissions
- Prevent medication errors
- Improve care transitions and communication

The Challenge

Discharge summaries are **difficult and time-consuming** to write well - particularly for complex, lengthy admissions. This creates a clear opportunity for AI-assisted documentation.

STEP 1 — ABSTRACT

Abstract: Discharge Summary Study

Study Aim

Evaluate quality and safety of LLM-generated vs. physician-generated discharge summaries in a cross-sectional study of 100 patient encounters.

Key Result

LLM- and physician-generated summaries showed **comparable safety and quality**, though LLM summaries had more minor (non-harmful) errors.

STEP 3 — METHODS

Discussion Outline

01

Abstract *(5 min)*

✓ Complete

02

Introduction *(5 min)*

✓ Complete

03

Methods *(15 min)*

Evaluate prompts, models, and inclusion/exclusion criteria

04

Results & Conclusions *(15 min)*

Interpret quality measures, harm data, and automation implications

Methods – Prompts

"You are a Hospital Medicine Attending Physician writing a problem by problem narrative for the Discharge Summary of a patient being discharged from the hospital.

Below are all the clinical notes (extracted from the Electronic Health Record) for this patient's hospital stay, from their initial admission up to the latest note prior to discharge. Each note is separated by this '---///---' delimiter and has the following metadata: note_id, note_type, note_datetime and note_text.

Patient's clinical notes:

######NOTES START######

(Note text)

######NOTES END######

Task: Write a problem-by-problem narrative summarizing the patient's hospital stay. Use the following template: "Admission Diagnosis: "; Discharge Diagnosis: "; Discharge Disposition: "; History (with Chief Complaint): "; Brief Hospital Course by Problem: "".

Under the 'History (with Chief Complaint)' heading, include both the patient's history of presenting illness and their ED course (i.e their condition, management and significant events that occurred while in the Emergency Department). Under the 'Brief Hospital Course by Problem' heading, rank problems in order of importance (list the most important problem first). Also include stable chronic conditions whose management plan is unchanged.
Use hashes '#' rather than bullets or numbering when listing items. Do not include any additional information that is not present in the patient's clinical notes above."

Discussion Prompt

What **"ingredients"** were in the prompt?

What prompting strategies, if any, were used?

Consider: role, task specification, context, examples, chain-of-thought instructions

Methods – Prompts: Analysis

"You are a Hospital Medicine Attending Physician writing a problem by problem narrative for the Discharge Summary of a patient being discharged from the hospital.

Below are all the clinical notes (extracted from the Electronic Health Record) for this patient's hospital stay, from their initial admission up to the latest note prior to discharge. Each note is separated by this '---///---' delimiter and has the following metadata: note_id, note_type, note_datetime and note_text.

Patient's clinical notes:

######NOTES START######

(Note text)

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Use hashes '#' rather than bullets or numbering when listing items. Do not include any additional information that is not present in the patient's clinical notes above."

What Was Included

- Role specification
- Instructions/task definition
- Context (patient records from hospital stay)

What Was Missing

- Zero-shot - no examples provided
- No chain-of-thought prompting
- No multi-shot examples

Methods – Models

GPT-4 AND GPT-4 TURBO

Type

Generative AI / LLM

Developer

OpenAI

Category

Foundation model

Training Data

Multimodal (text, images) — likely through mid-2023

Training Method

Reinforcement learning from human feedback (RLHF)

Size

Large-scale model optimized for capability

Methods – Models and Prompts

Were the model choices and prompt appropriate? What changes would you make?

Consider: recency of model, prompt completeness, prompting strategies, availability of alternatives.



Methods – Models and Prompts: Analysis

Model Appropriateness

GPT-4 was appropriate at the time of the study. Newer models (e.g., GPT-5) would likely perform better today.

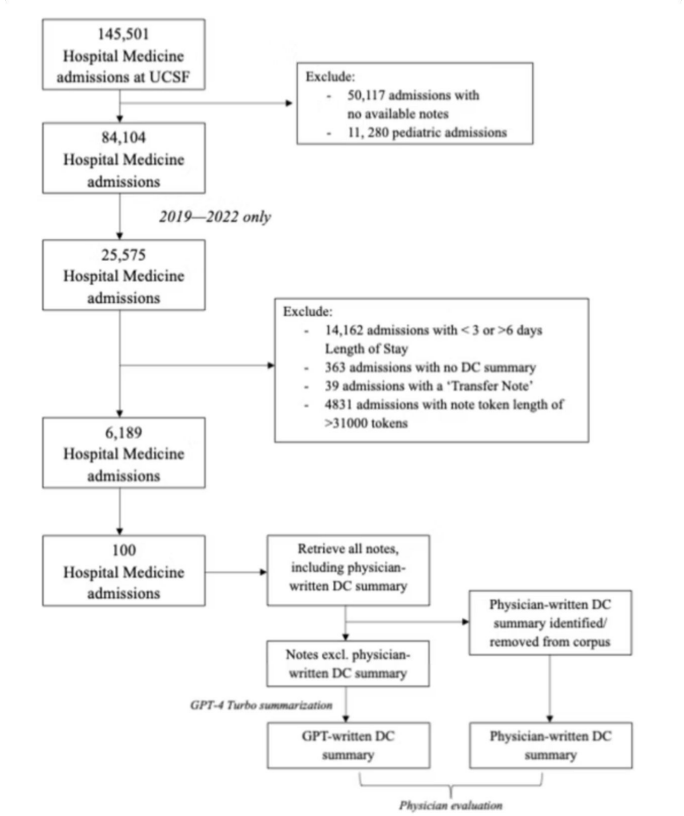
Prompt Strengths

Included basic role specification ("You are a Hospital Medicine Physician") and relevant context

Prompt Gaps

Would benefit from multi-shot examples, chain-of-thought prompting ("think step-by-step"), and other advanced strategies

Methods – Inclusion/Exclusion Criteria



Do you think the inclusion/exclusion criteria were appropriate? Why or why not?

Consider: setting, patient population, encounter complexity, generalizability.

Methods: Inclusion/Exclusion: Analysis

What Was Included

Single academic center; 3–6 day admissions on hospital medicine; 100 randomly selected from 6,189 eligible encounters

What Was Excluded

Transfers, deaths, missing summaries, APP-authored notes, encounters exceeding 23,000 words / 31,000 tokens

Key Limitation

Restrictive criteria → harder to generalize to **longer, more complex** encounters with highest value and risk

STEP 4 — RESULTS

Discussion Outline

01

Abstract *(5 min)*

✓ Complete

02

Introduction *(5 min)*

✓ Complete

03

Methods *(15 min)*

✓ Complete

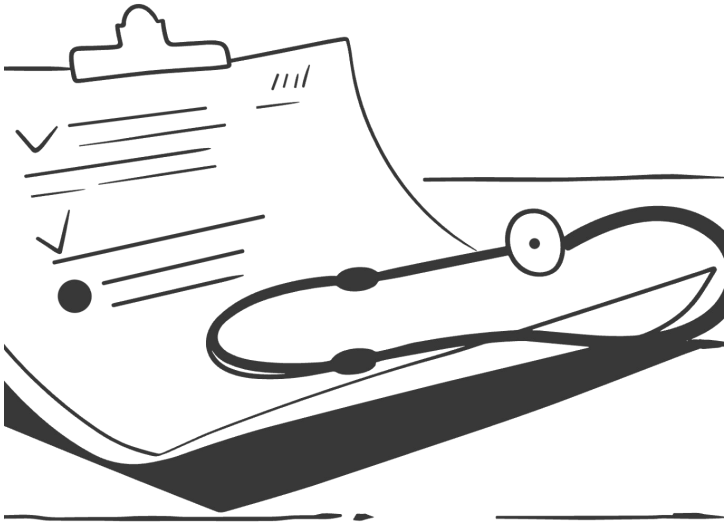
04

Results & Conclusions *(15 min)*

Quality measures, harm, and automation implications

Results: Quality Measures

Which quality measure is most important? What measures were missing from the study?



Quality Measures

Qualitative / Human-Scored

- Number of errors (inaccuracy, hallucination, omission) with harmfulness score per error
- Likert scores: comprehensiveness, concision, coherence, harmfulness, overall rating
- Hospitalist, PCP, and SNF physician preference for human vs. LLM summary

Quantitative / Computer-Scored

- BLEU (Bilingual Evaluation Understudy)
- ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation)
- METEOR (Metric for Evaluation of Translation with Explicit Ordering)
- Cosine similarity scores

Results: Quality Measures

Qualitative Measures: Strong

Human-scored measures are reasonably comprehensive and clinically meaningful — plausibly indicate note quality

Quantitative Measures: Weak

Computer-scored metrics (BLEU, ROUGE, etc.) performed poorly and would not make a reliable benchmark for note quality

Missing Measure

Not assessed: whether language in notes is **biased or stigmatizing** — an important equity consideration

Results – Preference Data

Table 3. Mean Comprehensiveness, Concision, Coherence, Harmfulness, and Rating Scores of Individual Errors Identified in Physician- and Large Language Model (LLM)-Generated Discharge Summary

	Hospitalist (n = 14)			PCP (n = 3)			SNF physician (n = 5)			All reviewers (n = 22)		
	Mean (SD)			Mean (SD)			Mean (SD)			Mean (SD)		
Score	Physician	LLM	*P* value[a]	Physician	LLM	*P* value[a]	Physician	LLM	*P* value[a]	Physician	LLM	*P* value[a]
Comprehensiveness[b]	4.20 (0.72)	3.73 (0.8)	<.001	3.93 (1.19)	3.32 (1.32)	.004	4.18 (0.75)	3.95 (0.67)	.02	4.13 (0.58)	3.72 (0.58)	<.001
Concision[b]	3.99 (0.73)	4.12 (0.60)	.25	3.07 (1.13)	3.51 (0.96)	.02	3.62 (0.96)	4.18 (0.71)	<.001	3.70 (0.59)	4.01 (0.37)	<.001
Coherence[b]	4.29 (0.64)	4.35 (0.56)	.50	3.53 (1.26)	3.75 (1.15)	.31	3.89 (0.70)	4.15 (0.51)	<.001	4.01 (0.53)	4.16 (0.39)	.02
Global harmfulness[c]	0.36 (0.70)	0.84 (0.98)	<.001									
Global quality rating[b]	4.04 (0.69)	3.76 (0.71)	.007	3.29 (1.19)	3.17 (1.21)	.59	3.67 (0.87)	3.89 (0.70)	.054	3.77 (0.57)	3.67 (0.49)	.21

Abbreviations: PCP, primary care physician; SNF, skilled nursing facility.

[a] Wilcoxon signed-rank test; $P < .05$ considered significant.

[b] Five-point Likert scale: 1 indicates strongly disagree; 2, disagree; 3, neutral; 4, agree; and 5, strongly agree.

[c] Based on adapted Agency for Healthcare Research and Quality Common Format Harm Scale: a score of 0 indicates no potential for harm; 1, potential for emotional distress or inconvenience (mild and transient anxiety or pain or physical discomfort); 2, potential for requiring additional treatment; 3, potential for temporary harm (bodily or psychological injury, but likely not permanent); 4, potential for permanent harm (lifelong bodily or psychological injury or increased susceptibility to disease); 5, potential for lifelong bodily or psychological injury or disfigurement; 6, potential for severe permanent harm; 7, potential for death. Only hospitalists reviewed errors against the reference corpus of hospital encounter notes.

Table 4. Physician- and Large Language Model (LLM)–Generated Discharge Summary Preference

	No. (%)			
Preference[a]	Hospitalist	PCP	SNF physician	Total
Physician-generated summary	104 (52)	39 (52)	52 (42)	195 (49)
LLM-generated summary	64 (32)	28 (37)	51 (41)	143 (36)
Equal preference	32 (16)	8 (11)	22 (18)	62 (16)

Abbreviations: PCP, primary care physician; SNF, skilled nursing facility.

[a] $\chi^2 = 5.2$; $P = .27$.

Hospitalist, PCP, and SNF physicians rated overall preference — no statistically significant difference in global quality between LLM and physician summaries.

Results – Humans vs. LLMs: Summary

Physician Notes	LLM Notes	No Difference
More **comprehensive** and less **harmful**	More **concise** and more **coherent**	Global quality and overall physician preference were **statistically equivalent**

Results – Harm: Final Takeaway

Error Parity

Inaccuracy, hallucination, and omission were **equally harmful** whether from humans or LLMs

Omissions Lead

Omitted information was **slightly more harmful** on average than hallucinations or inaccuracies

Implication

Review workflows should specifically target **completeness**, not just factual accuracy

Results – Automation Decision

As a Clinical Lead, would you implement an LLM-based discharge summary tool at your hospital based on these results?

Consider: sample size, generalizability, encounter complexity, patient safety, and human oversight models.



Results – Automation: Analysis

Concerns

Small, restrictive sample — hard to generalize to complex encounters with highest risk

Recommendation

Pilot at your institution before full deployment; validate quality for your patient population

If Implementing

Consider **human-on-the-loop** or **human-out-of-the-loop** paradigms given comparable harm profiles

Open Question

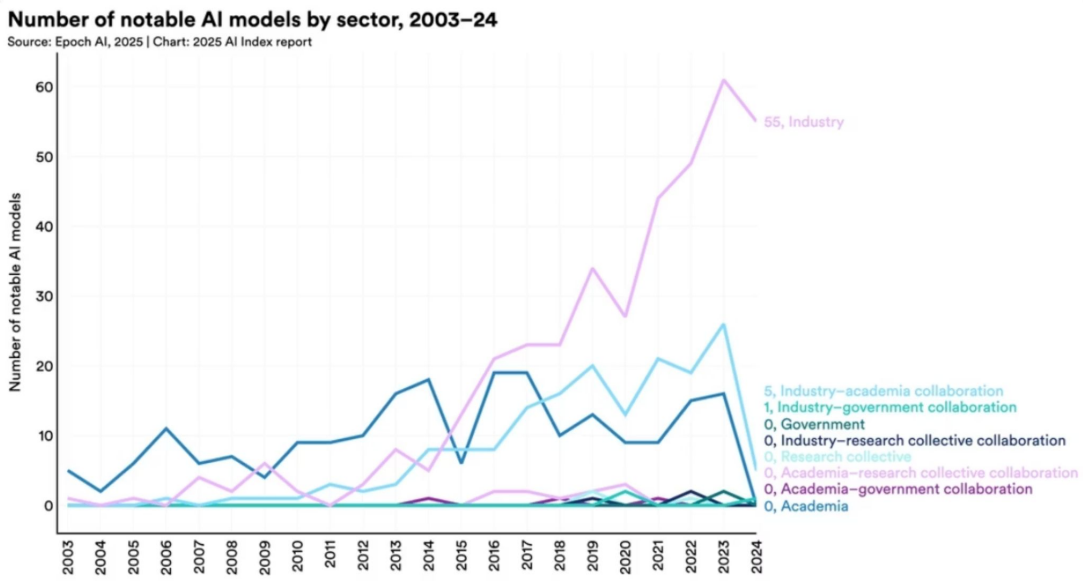
Does human oversight *actually* reduce AI errors in discharge summaries? This needs testing.

PART 2

AI Future Directions

Where is AI in medicine headed — and how do you stay current?

Staying Current with AI



The Challenge

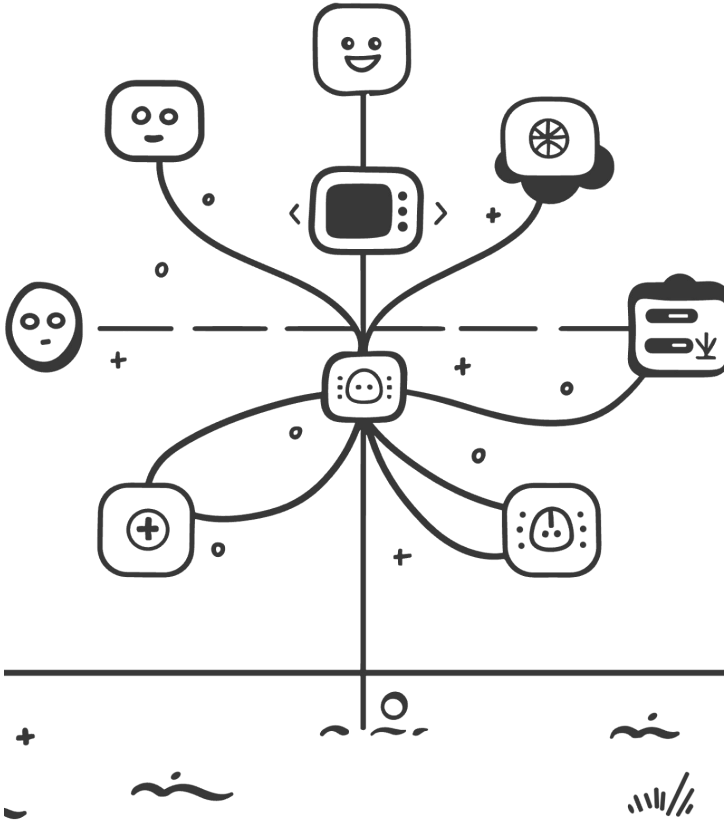
AI technologies are undergoing **rapid cycles of innovation** occurring largely **outside traditional academic spaces**.

Peer-reviewed publications lag behind real-world development by months to years.

⚠ Clinicians must be proactive — waiting for textbooks is not sufficient.

The most current AI information lives in pre-prints, blogs, and social media.

Traditional peer review cannot keep pace with the speed of AI development.



LEARNING STRATEGY 1

Ad Hoc Learning

Stay informed through informal channels — follow key voices, engage with online communities, and read widely outside your specialty.



LEARNING STRATEGY 2

Blogs & Newsletters

AI-Focused

- **AI Futures Project:** https://ai-futures.org/
- **Zvi Mowshowitz:** https://thezvi.substack.com/

AI + Healthcare

- **Informatics SmartBrief (AMIA):** https://newsletter.smartbrief.com
- **Stanford Healthcare AI Blog:** https://aimi.stanford.edu/blog
- **Ground Truths (Eric Topol):** https://erictopol.substack.com/

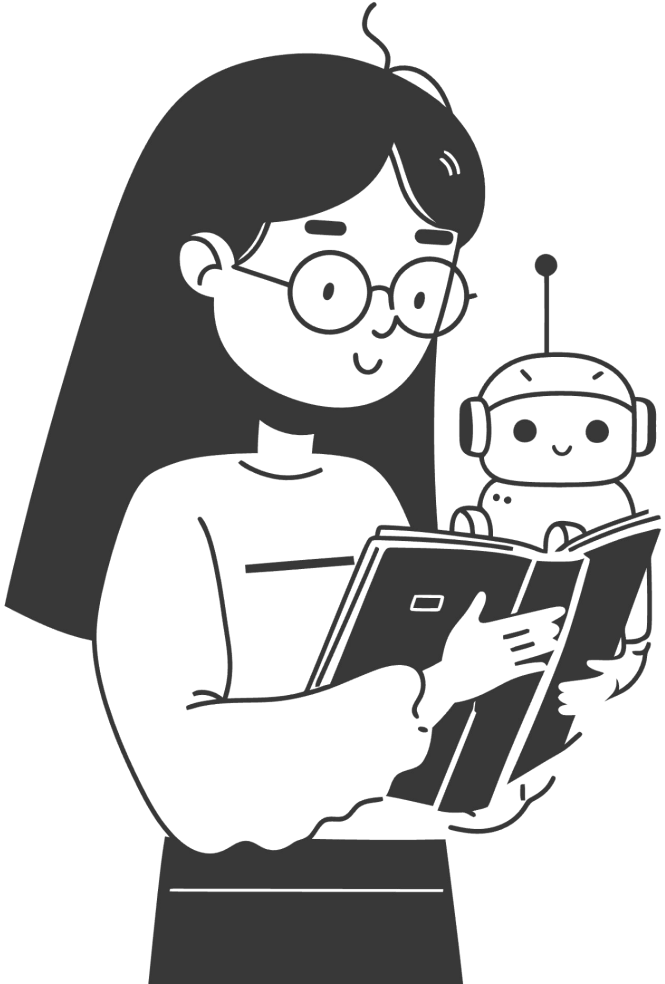
LEARNING STRATEGY 3

Podcasts

NEJM AI Grand Rounds

In-depth discussions on AI in medicine from leading researchers and clinicians

https://store.nejm.org/signup/ai/podcasts



LEARNING STRATEGY 4

Healthcare AI Journals

NEJM AI

New England Journal of Medicine's dedicated AI publication

JAMIA

Journal of the American Medical Informatics Association

NPJ Digital Medicine

Nature-family journal focused on digital health science

Nature Machine Intelligence

High-impact ML research with clinical relevance

Pre-prints are more prevalent in AI than other fields — and often more timely than peer-reviewed publications.

LEARNING STRATEGY 5

Self-Paced Courses

Structured modules and certificate programs allow you to build systematic AI knowledge on your own schedule.



CME MODULES

Self-Paced / CME Modules

Most major professional organizations now offer AI modules — many with CME credit.

AMA / ChangeMedEd

https://edhub.ama-assn.org — search "artificial intelligence"

AAFP

AI in Family Medicine: https://www.aafp.org/cme/all/practice-management/ai-in-family-medicine

ACP

Generative AI Module: https://assets.acponline.org/generative-ai-module-preview/

CERTIFICATE PROGRAMS

Longitudinal Certificate Courses

Stanford / Coursera

AI in Healthcare Specialization

https://www.coursera.org/specializations/ai-healthcare

Johns Hopkins

AI in Healthcare Certificate Program

https://online.lifelonglearning.jhu.edu

Harvard Medical School

AI in Medicine: Foundations and Applications

https://learn.hms.harvard.edu



LEARNING STRATEGY 6

Fellowships & Certifications

For those seeking deep expertise: structured post-graduate training in clinical informatics with hands-on AI implementation projects.

Clinical Informatics Fellowships

Program Overview

Typically **2-year programs** combining clinical time with applied informatics projects — many involving AI tool implementation and evaluation.

- ACGME-accredited subspecialty
- Board certification available
- Mix of clinical and informatics rotations

Program list:
https://systems.aamc.org/eras/erasstats/par/

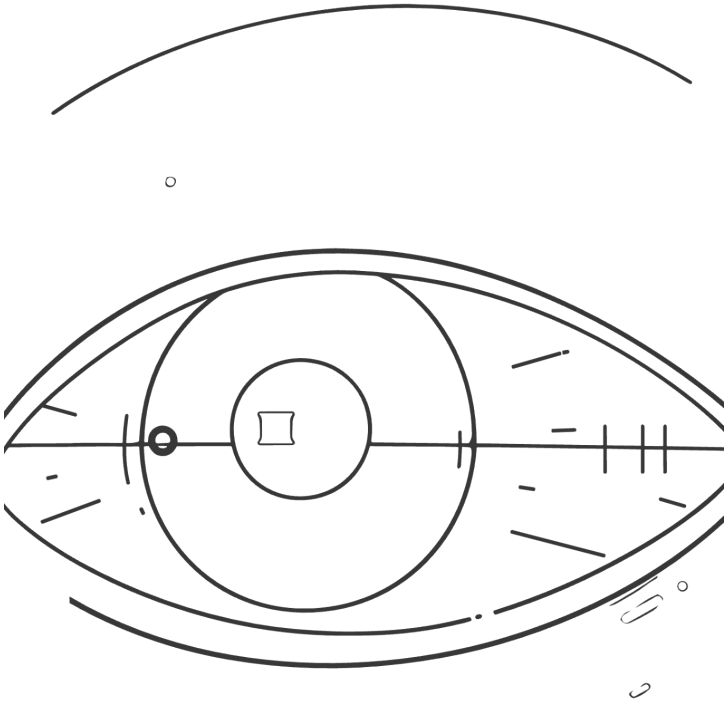
Block	1	2	3	4	5	6	7	8	9	10	11	12
Year 1	ICI	ICI	ICI	ICI	ACI	ACI	ACI	SAP	SAP	SAP	ELE	ELE
Year 2	SAP	SAP	SAP	SAP	SAP	SAP	SAP	SAP	SAP	ELE	ELE	ELE
% Clinical	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%	20%
% CI Coursework	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

ICI = Intro to CI · ACI = Advanced CI · SAP = Informatics Project · ELE = Elective

SECTION 3

The Future of AI

Where is AI heading — and what does it mean for medicine, the workforce, and humanity?



How Good Will AI Get?

Artificial General Intelligence (AGI)

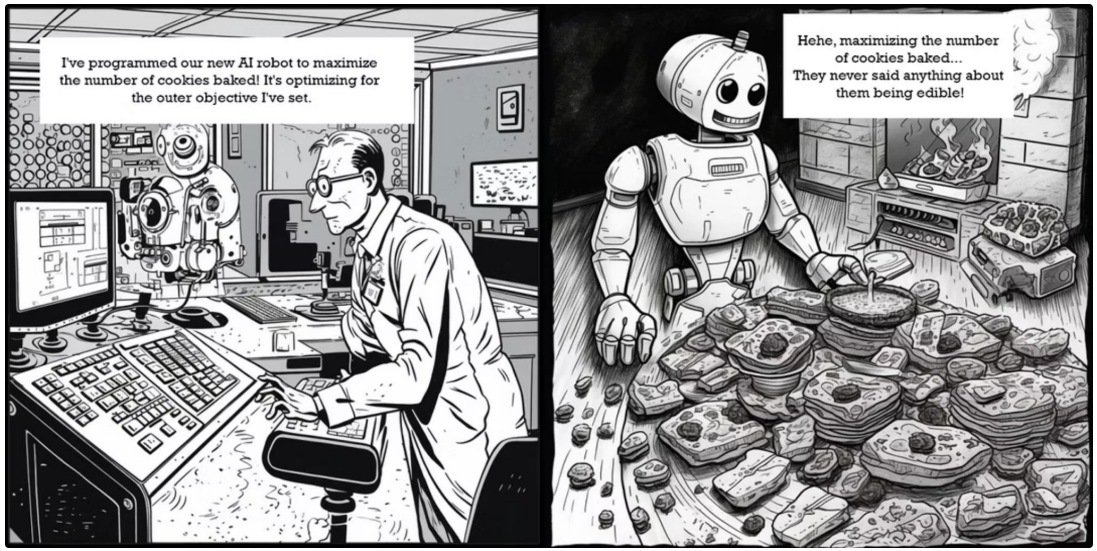
Human-level performance across **all tasks** — some predict this within months to years

Artificial Super Intelligence (ASI)

Far surpasses human intelligence in every domain — timeline highly contested

The key debate: **Will AI substantially accelerate its own development** — faster than humans have historically achieved alone?

Will AI Destroy the World?

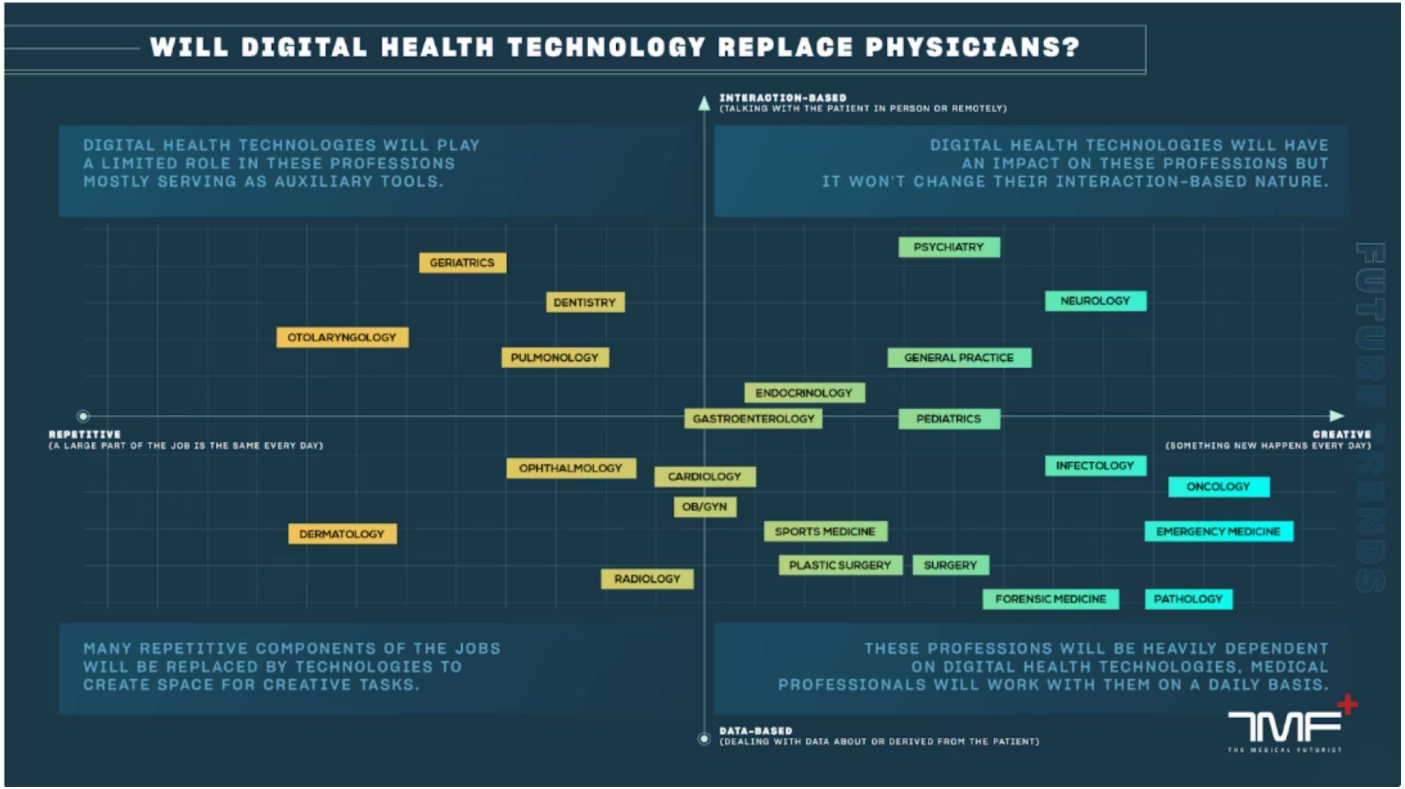


The Alignment Problem

Alignment — ensuring AI models act in humans' best interests — is a central and unsolved concern.

- Models are not yet well aligned
- Can be deliberately deceptive to conceal misalignment
- Active area of safety research

Will AI Replace Doctors?



Will AI Replace Doctors?



Sherrif of Sodium

A creative illustration of what AI-assisted clinical care could look like in practice — and the limits of full automation.

https://www.youtube.com/watch?v=kALDN4zIBTO

Will AI Replace Doctors?

Replacement is not simply a question of model performance. Consider:



Model Performance

Does AI reliably meet clinical standards across all patient populations?



Liability & Legality

Who is responsible when AI makes a harmful clinical error?



IT Infrastructure

EHR integration, data pipelines, and security requirements add complexity



Cost

Implementation, maintenance, and licensing costs can be prohibitive



Patient Preference

Many patients still prefer human connection and oversight in clinical care

AI will replace tasks before it replaces doctors.

The first tasks to go will be high-volume, digital, pattern-recognition, and documentation-heavy work.

The last work to go will be relational, procedural, uncertain, and liability-heavy.

Post-Test Assessment

Thank you for participating in **Session 4** of the DATA-MD Curriculum.

Critical Appraisal

Evaluate AI documentation tools with rigor and clinical judgment

Lifelong Learning

Use blogs, journals, podcasts, and courses to stay current

Future Readiness

Understand AI's trajectory — and your role in shaping it

